TakeSure — Judges' Q&A for the Discord judging round

Team Bug Hunters · Hack@BVP 8.0 · Every answer below is checked against the actual knowledge-base files and code (audited 18 Oct 2026). Say what is true; the honesty is the strength.

> **The one sentence to remember:** "Our verdict is only as good as the row it comes from, so every row shows its source and its grade, weak evidence can never produce more than AMBER, and an unknown pair is never shown as GREEN."

1. The data question: where does your data come from?

Q: This is a medical project. From where did you take the dataset? How sure are you that a RED is really RED and a GREEN is really GREEN?

Answer (say it in this order):

1. There is no ready-made dataset, so we built one. No public dataset of Indian herb × allopathic-drug interactions exists. Our knowledge base is hand-curated: 49 interaction rows today, covering 16 herbs, 34 drugs across 16 drug classes, and 28 Indian products mapped to their herbs. Every row was written by a team member reading the source, not scraped and not generated by AI.

2. Every row cites where it came from. 78 sources in total: 36 from reference databases (Memorial Sloan Kettering's "About Herbs", NIH LiverTox), 24 from monographs (WHO Monographs on Selected Medicinal Plants, the Ayurvedic Pharmacopoeia of India, Stockley's Herbal Medicines Interactions), 6 randomised controlled trials, 6 regulatory documents, 4 case series, 2 reviews. The source title, type, year and link or DOI are stored on the row and shown on the result page and in the PDF.

3. Every row carries an evidence grade, and we are honest about the mix. Grade A (human trials or systematic reviews): 2 rows. Grade B (case reports or small trials): 19 rows. Grade C (mechanism, animal or lab evidence only): 28 rows. More than half our rows are grade C, and we say so on screen. That is the real state of herb–drug science in India, and hiding it would be the dangerous thing.

4. The rules are built to be conservative because of that mix. A grade-C row can never produce more than AMBER, however scary the mechanism sounds. RED needs a major severity, or a moderate severity with a high-risk drug class (anticoagulants, antiplatelets, immunosuppressants, cardiac glycosides). Only 4 of our 49 rows are major. If the herb or drug is not in the table at all, the answer is "not in database", never GREEN. The worst pair always wins; a RED is never hidden by five GREENs.

5. So how sure are we? We are exactly as sure as the source we show you, and we tell you how sure that is. A RED for guggul with warfarin rests on a grade-B bleeding-risk row with Stockley's and MSKCC behind it; a GREEN for triphala with an oral medicine is a timing rule, not a claim of zero risk; an AMBER for ashwagandha with levothyroxine rests on two controlled trials that measured thyroid hormone changes. TakeSure never says "safe" in the absolute sense. It says: here is the published evidence, here is its grade, here are three questions for your doctor.

6. What we still have to do, and we say it before you ask. Only 6 sources carry a DOI today; the rest are database or monograph links. Every citation is flagged "needs verification" until a second reviewer re-opens it; each row stores the reviewer's name and the review date. Our Round 2 plan is a second-reviewer pass on all 49 rows and a review by a pharmacology or Ayurveda college faculty before any public release.

Q: Who are these sources? Why should we trust MSKCC or Stockley's?

MSKCC About Herbs is the Memorial Sloan Kettering Cancer Center's herb–drug reference, written by pharmacists and used by oncologists worldwide. NIH LiverTox is the US National Institutes of Health database on drug- and herb-induced liver injury. WHO Monographs on Selected Medicinal Plants are the World Health Organization's reviews of plant safety and pharmacology. The Ayurvedic Pharmacopoeia of India is the Government of India's official standard. Stockley's Herbal Medicines Interactions is the standard clinical reference on this exact topic, and we copied its evidence-grading model. These are the references a hospital pharmacist would open.

Q: Why not take the data from an API like Drugs.com or Medscape?

Two reasons. They do not cover herb × drug pairs for Indian products at all, so there is nothing to pull. And a licence-free API does not exist for this; copying their text would be a copyright problem. We cite public references and write our own rows.

Q: Why only 49 rows? Is that enough?

We chose depth over breadth on purpose: the 16 most-used herbs (ashwagandha, giloy, guggul, turmeric, triphala, mulethi, arjuna, brahmi and others) against the 16 drug classes people take long-term (thyroid, diabetes, blood thinners, BP, immunosuppressants, antiepileptics, statins, SSRIs and more). Each row takes about an hour of reading and writing. 49 rows cover the combinations that actually appear at a chemist's counter. Scaling to 200 is a curation workflow with a college partner, not a code problem.

Q: How do you decide the severity of a row?

From the source, not from us. If Stockley's or MSKCC classify an interaction as major, moderate or minor, we record that. If sources disagree, we keep the more cautious grading and mark the row for review. The rule engine then maps severity and grade to a colour with fixed rules R1 to R5; no human and no AI picks the colour at query time.

Q: What is "grade C with a plausible mechanism"?

A row where no human study exists but the pharmacology makes a clash plausible, for example giloy stimulating the immune system against a transplant drug that suppresses it. We show it as AMBER with the grade visible, which means "nobody has proven this in patients, but ask before you combine them". We never hide a grade-C row and never let it become RED on its own.

Q: What if the patient takes a product that contains ten herbs?

The products table maps each Indian product to its listed herbs (Chyawanprash → amla, ashwagandha and others). Each herb is checked against each medicine, and the worst row wins. If a product is not in our table, we ask the user to type the herbs or we say "not in database".

Q: Do you know the dose? Interactions depend on dose.

We do not model dose, and we say so. The verdict is "this pair has a published interaction of this severity"; the management text often mentions the dose context from the source, for example a trial at 600 mg a day. Dose-specific advice is the doctor's job, which is why the output is three questions for the doctor.

Q: Has a doctor validated your knowledge base?

Not yet, and we will not claim it. The rows are written from clinical references by the team; the second-reviewer flag is open on every row. The next step is a faculty review at an Ayurveda college or a pharmacology department, which is also our first ask of the mentors.

Q: What happens when a source is updated or retracted?

Every row has a knowledge-base version stamp and a reviewed-on date; the result and the PDF show the version. A correction is a new row version, and the "report incorrect information" button on every result creates a curation item on the dashboard.

2. "How accurate is it?" questions

Q: What is your accuracy number?

For the verdict there is no accuracy number to quote, because the verdict is a deterministic lookup: the same inputs always produce the same colour, and the colour is only as right as the row. The thing to measure is the row, and that is what source and grade are for. For OCR and name matching we have not published a measured figure; the confirmation screen is the control. Measuring them on a labelled set of real Indian boxes is a Round 2 deliverable.

Q: Could the system say GREEN for a dangerous pair?

Only if our table has a wrong minor row, which is a curation error, not a system error, and the source would be visible to catch it. Structurally: an unknown pair is GREY, not GREEN; a recognised pair with no row is GREEN but labelled "no known interaction in knowledge base version 1.0.0", which is a statement about our table, not about the world. The wording matters and we chose it deliberately.

Q: Could it say RED when it is actually fine, and scare people?

RED is reserved for major severity or moderate severity with a high-risk class, 4 major rows today. RED never says "stop"; it says "high risk, talk to your doctor before combining" and shows the source. False alarms are a real cost, which is why timing interactions like triphala are GREEN with a gap, not AMBER.

Q: What is the confidence of the OCR reading?

Tesseract returns a confidence per word, but we do not rely on it: every extracted name is shown back with its match score and type (exact, brand, synonym, fuzzy) and the user confirms. Below a 70 score the name is marked unrecognised and must be typed. Nothing enters the rule engine unconfirmed.

Q: What if the fuzzy matcher maps "Thyronorm" to the wrong drug?

Brand names are an exact dictionary hit (score 100, via "brand"), not fuzzy. Fuzzy matching only handles spelling errors, with a length guard, and anything between 70 and 85 is explicitly marked "please confirm". The review screen exists precisely so that a wrong match is caught by the user before any verdict.

3. Medical, safety and legal

Q: Is this medical advice? Who is liable if a patient is harmed?

It is decision support, and every screen and PDF says so. TakeSure never says "stop", "start" or a dose. The output is the published evidence, its grade and three questions for the doctor. The AI explainer is filtered for directive language in English and Hindi, and if it slips, the knowledge-base text is shown instead. Before public launch we would get a clinical review and a legal review; for the hackathon it is a prototype used with that disclaimer.

Q: Why not just tell the patient to stop the herb if it is RED?

Because we are not their doctor and we do not know their history. Stopping a herb can be wrong too; the doctor may adjust the drug instead. RED means "do not combine without talking to your doctor", and the three questions make that conversation happen.

Q: What about pregnancy, children, kidney disease?

Rows carry population flags (for example "pregnancy", "hyperthyroid history") shown as "extra care if". We do not take patient details beyond that; it is a deliberate privacy choice for the MVP and a stated limitation.

Q: Is Ayurveda being judged by allopathic standards here?

No. We do not rate Ayurvedic products as good or bad; we only flag documented interactions between a specific herb and a specific drug, citing Ayurvedic Pharmacopoeia and WHO plant monographs alongside clinical references. Many rows are GREEN or timing-only. The goal is that patients can keep using both systems safely.

Q: Are you a registered medical device?

No, and we do not need to be for a prototype that gives referenced information and questions. If TakeSure ever gave dosing or diagnosis it would fall under software-as-medical-device rules; we have scoped it to stay outside that on purpose.

Q: What data do you store? Is it private?

Photos are deleted after text extraction. We store entity ids, verdict, language, optional state and a timestamp: no names, no phone numbers, no IP addresses. ADR reports are stored only with the reporter's consent flag. The dashboard is built from that anonymised log.

4. Technical

Q: Which model did you train?

None, deliberately. Pre-trained Tesseract reads the photo, a hosted LLM rewords the matched row, and deterministic rules over our cited knowledge base decide. What we built is the data, the bridge dictionary, the rules, the guardrails and 51 tests. Training a classifier would make the verdict unexplainable and there is no labelled dataset to train it on.

Q: How do you stop the LLM from hallucinating?

It only receives the matched rows, never the open web; temperature 0; a system prompt that forbids adding facts; the output must match a strict JSON schema; a filter rejects "stop taking", "increase the dose", any mg figure, in English and Hindi. If anything fails, the knowledge-base text is shown. The verdict is computed before the LLM runs and cannot be changed by it.

Q: Why Groq and which model?

Low latency and a free developer tier; the current model is the open-weight gpt-oss-120b. The provider is swappable through configuration (Anthropic, Gemini, any OpenAI-compatible endpoint). The explainer is optional: with no key the product still works.

Q: Why SQLite? Why not a vector database?

The knowledge base is small, relational and read-heavy; after matching, lookups are exact ids, so a relational store is correct and a vector search would add fuzziness exactly where we want none. SQLite for the laptop demo, PostgreSQL on Supabase in production, same code.

Q: Does it work offline? What needs internet?

OCR, matching, verdict, sources, PDF, ADR form and dashboard all run on one laptop with no internet. Only the AI rewording needs the Groq API, and it falls back automatically.

Q: How is it tested?

51 automated tests, run offline: 15 on the rule engine (every rule and the GREY cases), 10 on extraction and matching thresholds, 8 on LLM validation and fallback, 8 on knowledge-base integrity and demo scenarios, 10 on uploads and PDFs. Plus the end-to-end journey verified in the browser.

Q: How does Hindi work end to end?

UI strings from our own dictionary, Hindi OCR language file loaded, the explainer returns a Hindi summary, and the PDF uses a Devanagari font. The knowledge-base rows are written in English with a Hindi patient summary field.

Q: What is the architecture in one breath?

Next.js web app → FastAPI → Pillow and Tesseract for OCR → RapidFuzz over the bridge dictionary → rule engine over SQLite/PostgreSQL → Groq explainer with guardrails → ReportLab PDFs, ADR form and an analytics dashboard.

5. Product, users and impact

Q: Who is the user, really? Will a village patient use this?

First users are caregivers with smartphones managing an elder's medicines, and chemists at the counter. The chemist QR on the bill is the distribution channel to patients who would never install an app. Hindi, big verdict colour and three questions are designed for low-literacy use.

Q: Why would a doctor trust a patient's printout from an app?

Because the printout is a list of references with grades, not an opinion. It reads like a pharmacist's note: pair, severity, mechanism, management, sources. Doctors already use MSKCC and Stockley's; we hand them the same references.

Q: How does this help the Ministry of Ayush?

Its pharmacovigilance programme has received roughly 1,473 reports since 2018. We produce suspected-reaction reports in the official field format for the Ayush Suraksha portal, and anonymised co-use analytics by state. We link to the official portal rather than building a parallel system.

Q: What is the business model?

Free for citizens; API per query or subscription for pharmacy chains, hospital systems and telemedicine; programmes and CSR with the Ministry and Ayurvedic brands that want a trust layer; long-term ABDM integration so the verdict sits on the ABHA record.

Q: What stops a big player copying you?

The curated, graded, Indian-brand-aware knowledge base is the moat; it is domain time, not code. We plan to keep it open and co-curated with Ayurveda colleges, which makes TakeSure the reference others use rather than a competitor they copy.

Q: Where do the 37 %, 65 %, 19 %, 75 % and 1,473 come from?

37 % and 65 % from an Indian hospital study of patients' herbal-medicine use (Bhat et al., Trop J Pharm Res 2019); 19 % disclosure from an Indian J Med Res editorial on herb–drug interactions (PMC4643608); 75 % of herbal users with a side effect who never told the doctor, same study family; 1,473 ASU&H reports from Ministry of Ayush pharmacovigilance figures. All are on the references slide with links.

6. Team, process and the hackathon

Q: What did you build before and what will you build on site?

Before: the research, the 49 knowledge-base rows, the design, and a prototype proving the pipeline end to end. On site: rebuild the application per the rules and extend it: 100-plus rows with a second reviewer on every citation, OCR hardening on real boxes and handwritten slips, Hindi in every PDF, the chemist QR flow, and a public deployment. The research is the reusable asset.

Q: Who did what?

Dhriti: backend, rule engine, medical research. Shiwani: frontend, Hindi/English UI, result card, demo and pitch. Khushi: knowledge base and evidence grading, OCR, PDF and ADR export. Everyone wrote knowledge-base rows.

Q: What was the hardest part?

Grading evidence honestly. It is tempting to call everything moderate to look thorough; we forced ourselves to record what the source actually says, which is why 28 rows are grade C and only 4 are major.

Q: What would you do with a mentor's help right now?

Two things: an introduction to an Ayurveda college or pharmacology faculty to review the knowledge base, and advice on whether an ABDM sandbox is realistic for the offline round.

Q: Why should you go to the next round?

Because the hard part, the curated and cited knowledge base, already exists and is auditable, and the prototype works end to end today. Our 24 hours on site go into making it bigger and public, not into making it exist.

7. Trap questions and how to answer without losing credibility

Trap	Answer
"So your data is basically copied from MSKCC."	"Our rows cite MSKCC the way a textbook cites a paper: we read it, wrote our own row, graded it, and added Indian brand and synonym mapping that MSKCC does not have. The citation is the point."
"Most of your rows are grade C. That is weak."	"Yes, and that is the honest state of the science. Our contribution is to show the weakness instead of hiding it: a grade-C row can never produce more than AMBER, and the grade is on the screen."

"A doctor would never trust an app."	"A doctor trusts references. We hand them a referenced note, not a verdict from a black box, and the questions are addressed to the doctor, not around them."
"Why is there no machine learning?"	"Because a trained classifier cannot be audited and there is no dataset to train it on. We use AI where it is safe, reading and rewording, and keep it out of the decision."
"What if your source is wrong?"	"Then the row is wrong and the source is visible to catch it. Every row has a version, a reviewer and a date; the feedback button creates a curation item; the second-reviewer pass is our next milestone."
"Can you prove it works on real prescriptions?"	"On printed labels and typed slips, yes, live. On handwriting it is weak and we say so; confirmation and manual entry are the controls, and measuring OCR on real boxes is on our Round 2 list."
"Isn't 'not in database' just a way of avoiding answers?"	"It is the most important answer we give. A tool that guesses when it does not know is the one that hurts people. 'Not in database' also logs the pair for curation, so the gap closes."

8. Numbers to memorise

Item	Value
Knowledge base	49 rows · 16 herbs · 34 drugs · 16 drug classes · 28 products · version 1.0.0
Severity split	4 major · 23 moderate · 22 minor
Evidence grades	2 A · 19 B · 28 C
Interaction types	32 pharmacodynamic · 10 pharmacokinetic · 5 timing · 2 contamination
Sources	78 total: 36 database (MSKCC, LiverTox) · 24 monograph (WHO, Pharmacopoeia, Stockley's) · 6 RCT · 6 regulatory · 4 case series · 2 review · 6 with DOI
High-risk classes (R5 escalation)	anticoagulant · antiplatelet · immunosuppressant · cardiac glycoside
Rules	R1 no row → GREEN · R2 minor → GREEN · R3 moderate or grade C → AMBER · R4 major → RED · R5 moderate + high-risk class → RED · unknown → GREY
Matching thresholds	≥ 85 accept · 70–85 confirm · < 70 unrecognised
Tests	51, offline
Statistics quoted	37 % co-use · 65 % lack knowledge · 19 % disclose · 75 % never told doctor about a side effect · 1,473 ASU&H reports since 2018

> **Do not say:** "clinically validated", "doctor-approved", "99 % accurate", "AI-powered diagnosis", "safe to take". **Do say:** "referenced", "graded", "decision support", "ask your doctor", "not in database".